

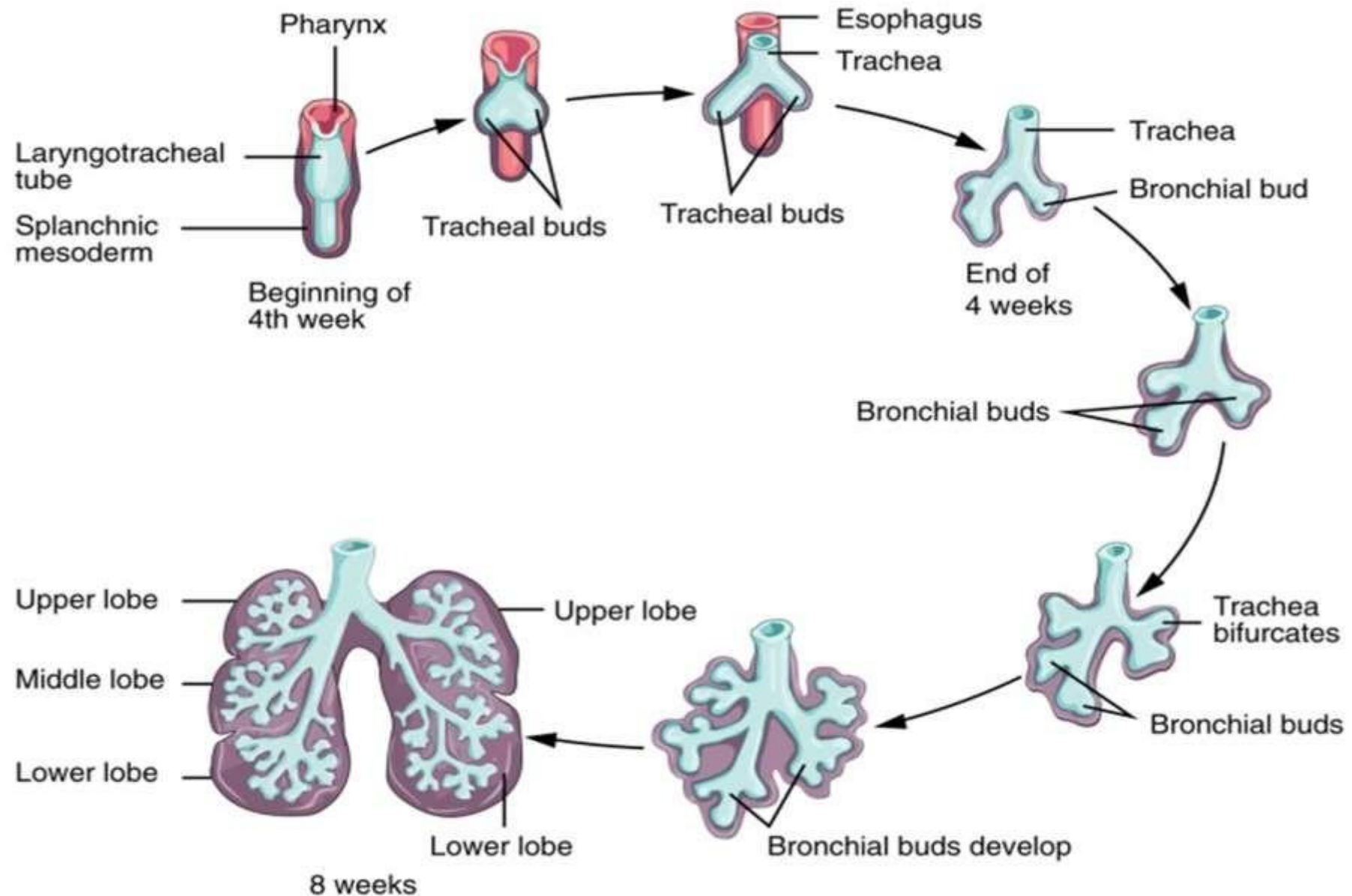
NEONATAL ACUTE RESPIRATORY DISTRESS SYNDROME

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Definition

- ❖ Acute lung disease of the newborn caused by surfactant deficiency.
- ❖ RDS is the clinical expression of surfactant deficiency and its histologic counterpart, Hyaline Membrane Disease (HMD).

Development of Lungs



Surfactant

- ❑ Surfactant is identifiable in fetal lung as early as 16 weeks, though its proper secretion begins after 24 weeks gestation and is synthesized most abundantly after the 35th week of gestation.
- ❑ Pulmonary Surfactants are phospholipids synthesized in the type II cells lining the alveoli surfactant.
 - Phospholipid produced by alveolar type II cells. Lowers surface tension.
 - As alveoli radius decreases, surfactant's ability to lower surface tension increases.
 - The half-life of surfactant is 30 hours

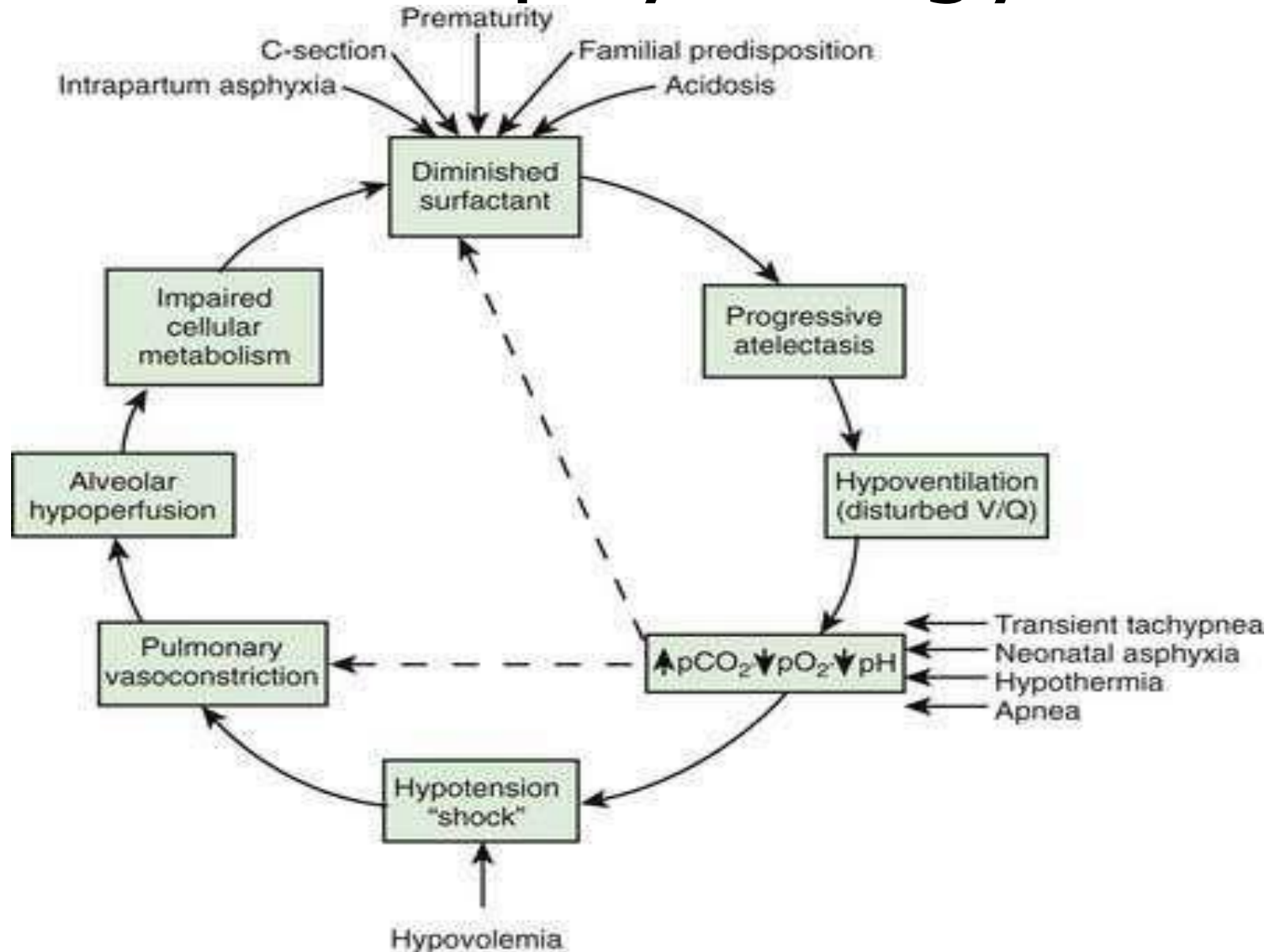
Function of the Surfactant

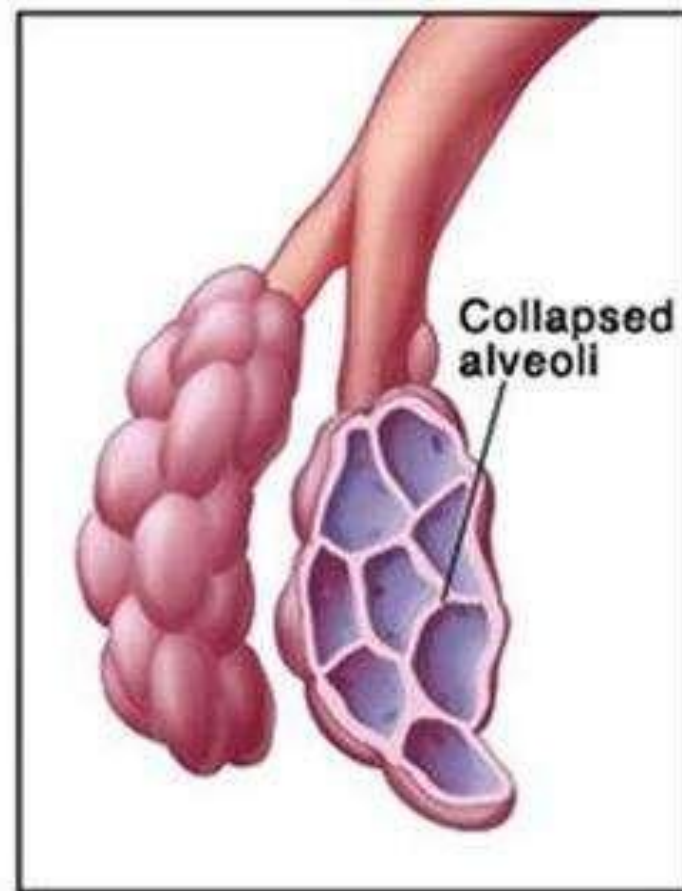
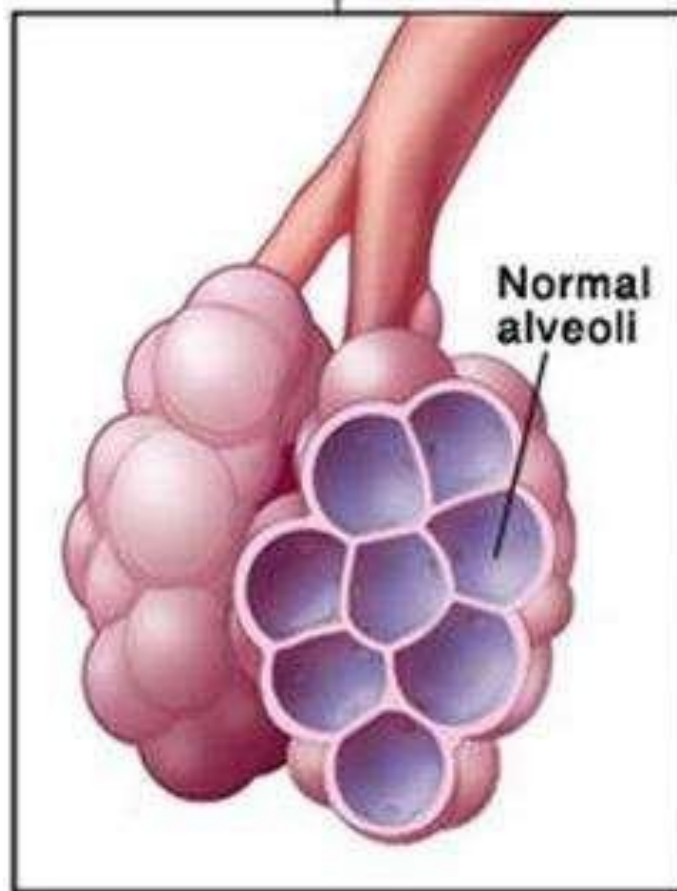
- ❖ Decrease the surface tension
- ❖ To promote lung expansion during inspiration.
- ❖ To prevent alveolar collapse and loss of lung volume at the end of expiration.

Causes of surfactant deficiency

- ❖ Pulmonary infections e.g. group B Strep
- ❖ Pulmonary hemorrhage
- ❖ Meconium aspiration pneumonia
- ❖ O₂ toxicity; barotrauma or volutrauma to the lungs.
- ❖ Congenital diaphragmatic hernia and pulmonary hypoplasia

Pathophysiology





Risk factors

- ✓ Prematurity
- ✓ Multiple births
- ✓ Maternal diabetes,
- ✓ C-section without labor
- ✓ Perinatal asphyxia
- ✓ Cold stress
- ✓ White race, male sex
- ✓ Hypothermia, hypothyroidism
- ✓ 2nd twin

Clinical Manifestations

- ❖ Appear within minutes of birth may not be recognized for several hours in larger preterm.
 - Tachypnea (>60 breaths/min),
 - Nasal flaring,
 - subcostal and intercostal retractions,
 - cyanosis & expiratory grunting.
- ❖ Breath sounds may be normal or diminished and fine rales may be heard.
- ❖ Progressive worsening of cyanosis & dyspnea.
- ❖ Cyanosis and pallor increase & grunting decreases.
- ❖ Apnea and irregular respirations are ominous signs.
- ❖ In most cases, symptoms and signs reach a peak within 3 days, after which improvement occurs gradually.

Diagnosis

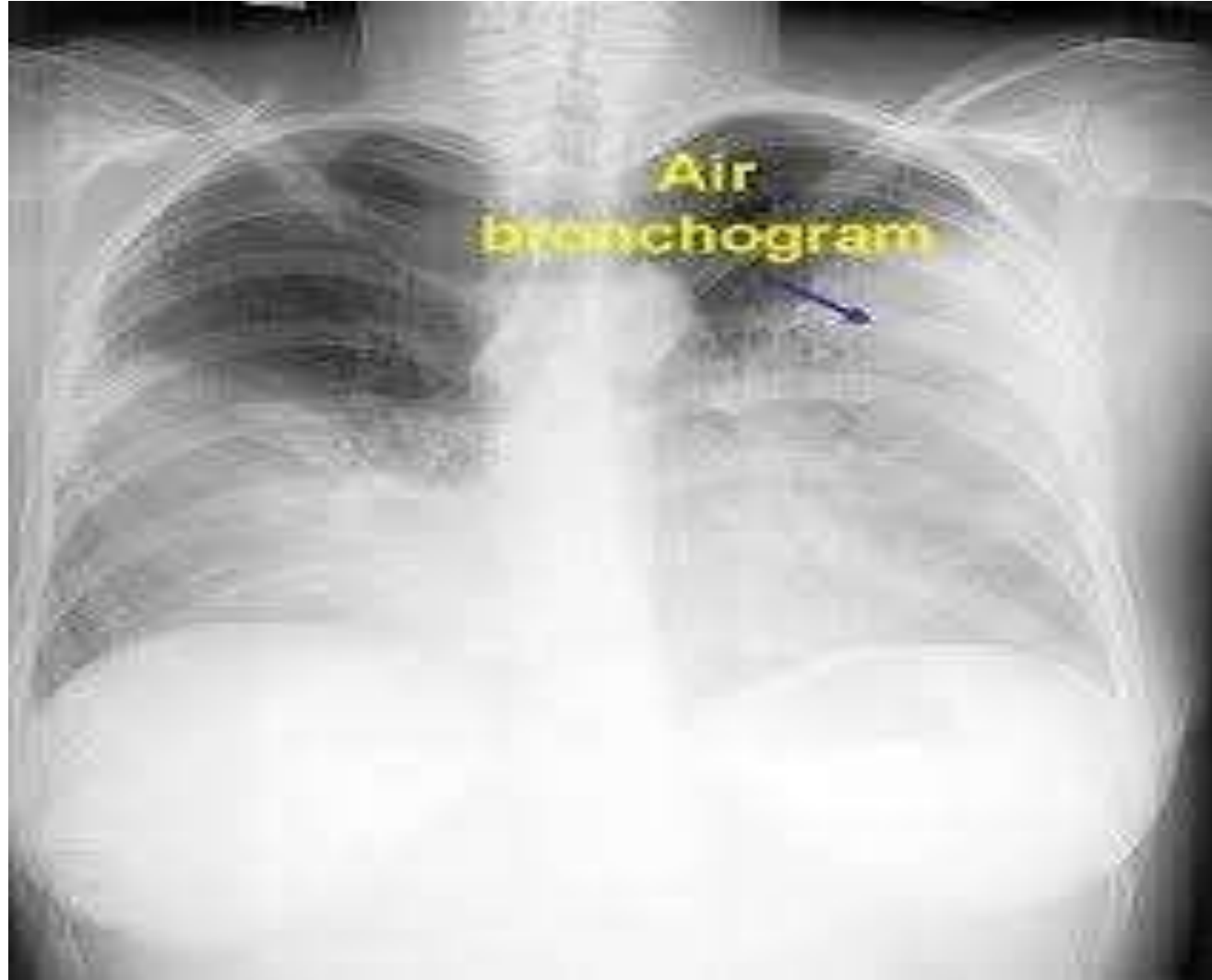
Chestx-ray:

Grade 1 (mild cases): The lungs show **fine homogenous ground glass shadowing**.



Shows air bronchograms and reticulonodular shadowing throughout the lung fields (often termed 'ground glass' appearance)

Grade 2: Widespread air bronchogram become visible.



Grade 3: Confluent alveolar shadowing.



Grade 4: Complete white lung fields with obscuring of the cardiac shadow.



Investigations

❑ ***Blood gas analysis:***

- Shows low oxygen and excess acid in the body fluids.

❑ ***Laboratory findings***

- are initially characterized by hypoxemia and later by progressive hypoxemia, hypercapnia, and variable metabolic acidosis.

Progression of the disease

Acute phase

- ✦ First 48-72 hr. after birth, newborn begins to have tachypnea, chest tightness.
- ✦ The sign and symptoms are peak on day 1-2.
- ✦ Patients may die if they did not receive adequate treatment.

Recovery phase

- ✦ Start on day 3-5
- ✦ Type II Pneumocytes are regenerated
- ✦ Decrease oxygen requirement

Management

Antenatal corticosteroid therapy:

Betamethasone 12 mg IM for 2 doses, 24 hrs apart, or Dexamethasone 6 mg IM for 4 doses, 12 hrs apart.

- ✓ They induce surfactant production and accelerate fetal lung maturation.
- ✓ Are indicated in pregnant women 24-34 weeks' gestation
- ✓ Optimal benefit begins 24 hrs after initiation of therapy and lasts seven days.
 - Early surfactant therapy: prophylactic use of surfactant in preterm newborn <27 weeks' gestation.
 - Early CPAP administration in the delivery room.

Treatment

Supportive treatment:

- Body temperature -
 - ✓ Scheduled “touch times” to avoid hypothermia and minimize oxygen consumption.
 - ✓ Placed in an isolette or radiant warmer to maintain core temperature between 37 ± 0.5 °C.

Nutritional support -

- ❖ For the 1st 24 hour, **10%DW** should be infused through a peripheral vein at a rate of 65–75 mL/kg/day.
- ❖ For VLBW and ELBW, TPN should be added Day 2-3, Na 3-4 mEq/kg/day and K 2-3 mEq/kg/day should be added
- ❖ Give a blood transfer when Hct is less than 40%

Oxygen therapy

- ✓ **Warm humidified oxygen** should be provided at a concentration initially sufficient to keep
 - PaO₂ 50-80 mmHg, pH 7.25-7.45, PaCO₂ 40-50 mmHg and SpO₂ 90–95%.
- ✓ To maintain normal tissue oxygenation while minimizing the risk of oxygen toxicity.
- ✓ O₂ box is not recommended for newborn with VLBW and ELBW because of high concentration of O₂ may increase risk of ROP (Retinopathy of prematurity).

Cont

- ✓ If the PaO_2 cannot be maintained above 50 mmHg at inspired oxygen concentrations of 60% or greater, applying CPAP at a pressure of 5–10 cm H₂O by nasal prongs (CPAP prevents collapse of surfactant-deficient alveoli, improves ventilation-perfusion matching).
- ✓ The amount of CPAP required usually decreases abruptly at about 72 hours of age, and infants can be weaned from CPAP shortly thereafter.

Complications of CPAP

- **Pulmonary air leaks** - over distension of the lungs caused by inappropriately high pressures
- **Decreased cardiac output** due to reduction in the venous return, decreased right ventricular stroke volume
- Gastric distension and '**CPAP belly syndrome**'
- Nasal irritation, damage to the septal mucosa, or skin damage and necrosis from the fixing devices.

Mechanical ventilation

Continue positive airway pressure (CPAP) is being use with 4-8 cm H₂O.

- ✓ To make Functional residual capacity (FRC) for the lung to prevent atelectasis.
- ✓ Usually started with 5 cm H₂O and increased by 1 cm H₂O in subsequent with increase oxygen by 10%.

Initial settings:

- Continuous flow, pressure-limited, ventilator conventional.
- Positive End-Expiratory Pressure (PEEP) 4-5 cm H₂O
- Frequency 40-60/min
- FiO₂ 50-60%

Cont...

Routes of administration:

- ✓ Nasal prongs
- ✓ Nasopharyngeal tube

Indication for ventilator

- ✓ Apnea with no improvement
- ✓ Cyanosis or $\text{PaO}_2 \leq 40 \text{ mmHg}$ (when using CPAP and high oxygen concentration)
- ✓ Signs of Respiratory failure
 - ✧ $\text{PaCO}_2 > 60 \text{ mmHg}$
 - ✧ Metabolic acidosis

Surfactant replacement therapy

Surfactant replacement therapy can reduce mortality and incidence of Chronic pulmonary disease.

There are 2 types of surfactant :

1. **Natural surfactant extract**

- ✦ Bovine(Survanta), Porcine(Curosurf)
- ✦ Natural surfactants appear to be superior, perhaps of their surfactant-associated protein content.
- ✦ Natural surfactants have a more rapid onset and are associated with a lower risk of pneumothorax and improved survival.

Cont...

2. **Synthetic surfactant**

- ✦ Exosurf and ALEC (Artificial Lung Expanding Compound)

3. **Newer surfactant**

- Synthetic surfactants with synthetic peptides modelled on surfactant proteins, Aerosolized surfactants.

Dose:

- Survanta 100mg/kg for the first and subsequent doses.
- Curosurf 200mg/kg for the first dose and 100mg/kg for the subsequent doses or 100mg/kg for all the doses.

Cont...

The main indications :

Prophylactic treatment:

- ✦ Being use for infant delivered during 23-29 wk of gestation and birth weight 600-1250 g
- ✦ Neonates with gestation < 30 weeks of gestation.
- ✦ Surfactant given within 15 minutes of birth before a diagnosis of RDS is made.

✦ Results :

- Improve dyspnea in first 48-72 hr of life (Decrease O₂ requirement, ventilation improved)
- Decreased incidence of pneumothorax
- Decrease mortality

Complications of surfactants:

- Transient hypoxia, bradycardia and fluctuating BP
- Rapid changes in lung compliance leading to barotrauma if not monitored.
- Pulmonary hemorrhage - more with natural(5-6%) as compared to synthetic (1 -3%).

Pharmacotherapy- beyond surfactant:

Nitric oxide

- Inhaled nitric oxide (iNO)– a selective pulmonary vasodilator improves oxygenation in preterm infants with severe RDS.
- Nitric oxide may be a signaling molecule in parenchymal lung growth & may reduce lung injury and chronic lung disease.

Thanks!